디지털 논리 회로 설계

(VHDL) Assignment-2

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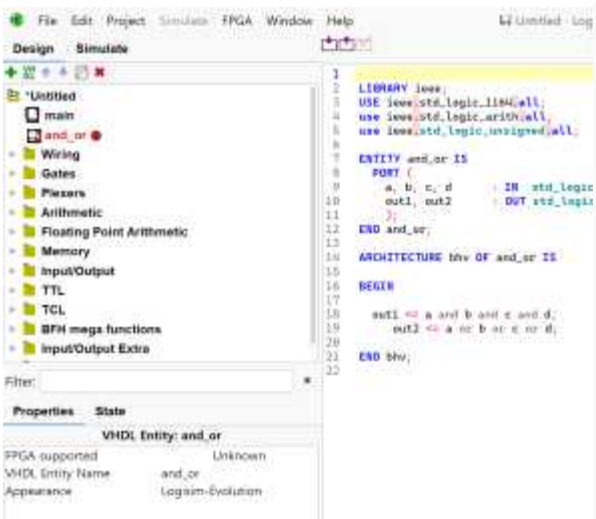
과제

1, Write VHDL code.
1) out1 <= a and b and c and d;
2) out2 <= a or b or c or d;
2, 8-to-1 Multiplexer
3, Use digital tubes to display letters: a,b,c,d,f,g
4, 25-persion voting machine
5, Design a VHDL module that counts the number of '1's in an input signal `vt` (a 13-bit vector) and displays the corresponding number or letter on a 7-segment display. The specific requirements are as follows:
- When the number of '1's in `vt` is between 0 and 9, display the corresponding digit (0-9).
- When the number of '1's in `vt` is between 10 and 15, display the letters A, B, C, D, E, F, respectively.
- The module should include an enable signal `en`. When `en` is '0', the display should turn off. When `en` is '1', the display should show the result.

1. and_or circuit

(1) VHDL code (and_or.vhd) & (2) Truth table

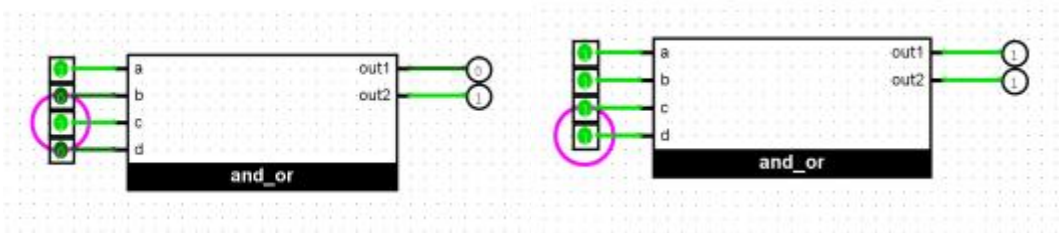
- Input: a, b, c, d
- Output1: abcd
- Output2: a+b+c+d



16 of 16 rows shown

a	b	c	d	x	y
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	0	1

(3) Entity & Simulation (and_or.circ)



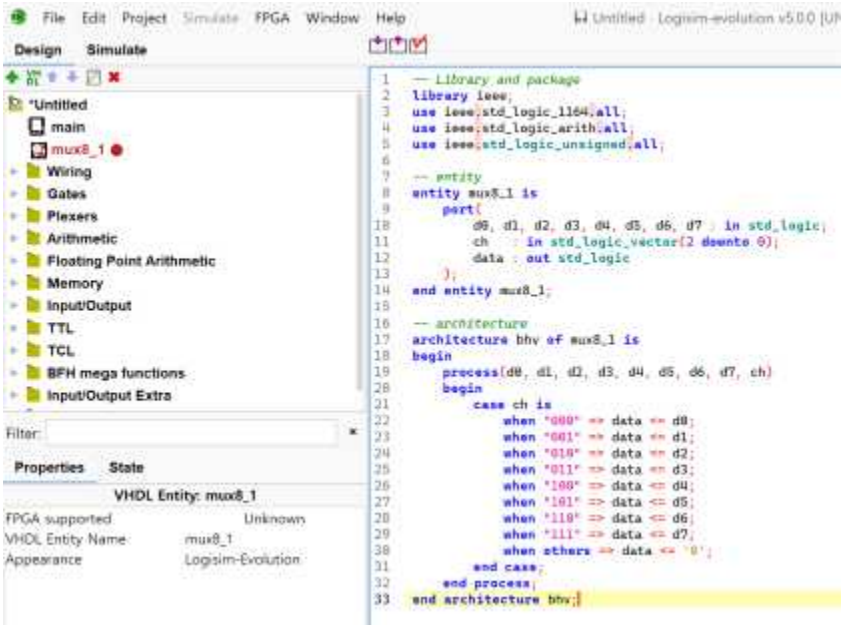
2. 8-to-1 Multiplexer (mux8_1)

- 개요: 8개의 1비트 입력(d0~d7) 중 3비트 선택 신호(ch)에 해당하는 입력을 출력(data)으로 전달

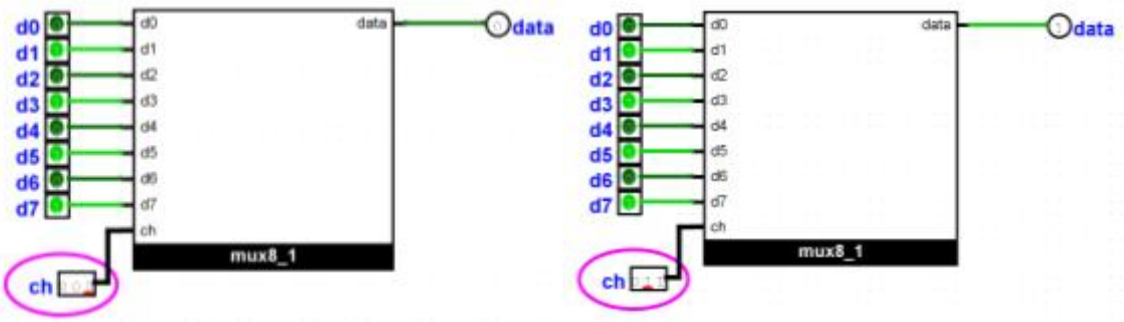
* multiplexer: 다수의 데이터 입력 신호 중 제어(선택) 신호에 의해 지정된 하나의 신호만을 출력

- Input1: 8개의 1비트 데이터 입력(d0 ~ d7)
- Input2: 3비트 선택 제어 신호(ch[2:0])
- Output: data = 선택 값("000" ~ "111")에 대응하는 입력값을 출력

(1) VHDL code (mux8_1.vhd)



(2) Entity & Simulation (mux8_1.circ)

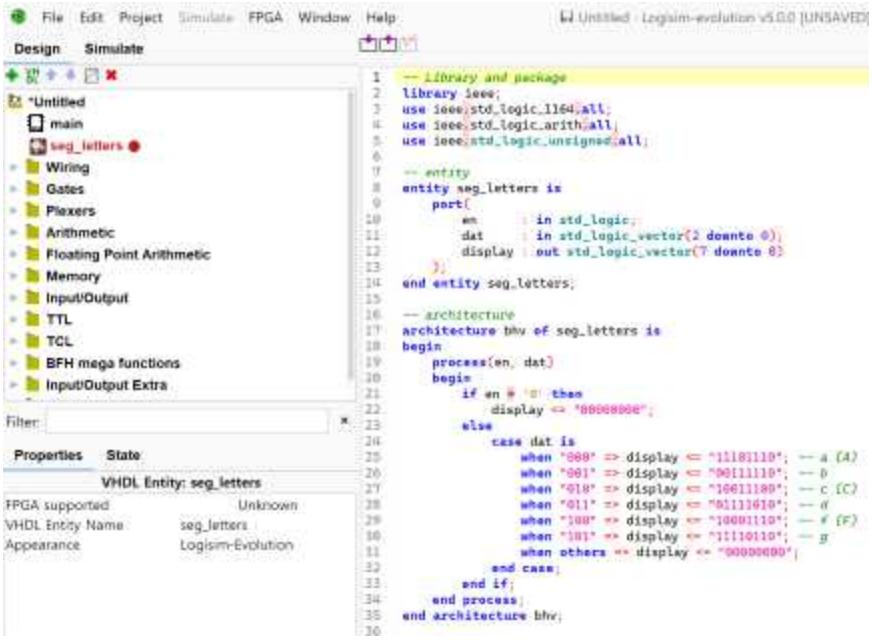


- 시뮬레이션에서 선택 신호를 000부터 111까지 변경하며 각 입력이 올바르게 선택되는지 확인함

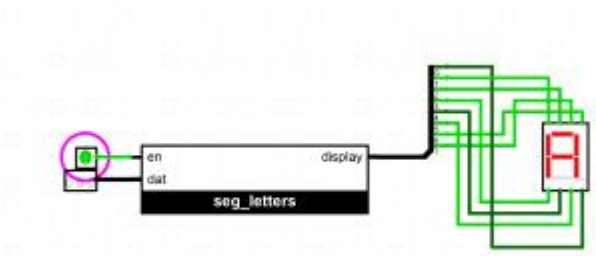
3. 7-Segment Display - letters (seg_letters)

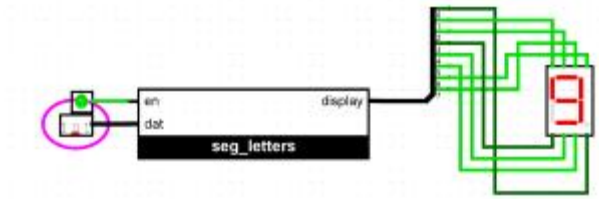
- 7-Segment Display에 문자 a(A), b, c(C), d, f(F), g 출력
- en=0일 때는 display 표시 X, en=1일 때 표시 O
- 출력 포트 display(7 downto 0)의 각 비트는 상위 비트부터 순서대로 세그먼트 (a, b, c, d, e, f, g, dp)에 대응

(1) VHDL code (seg_letters.vhd)



(2) Entity & Simulation (seg_letters.circ)





4. 25-persion voting machine (vote25)

- Input

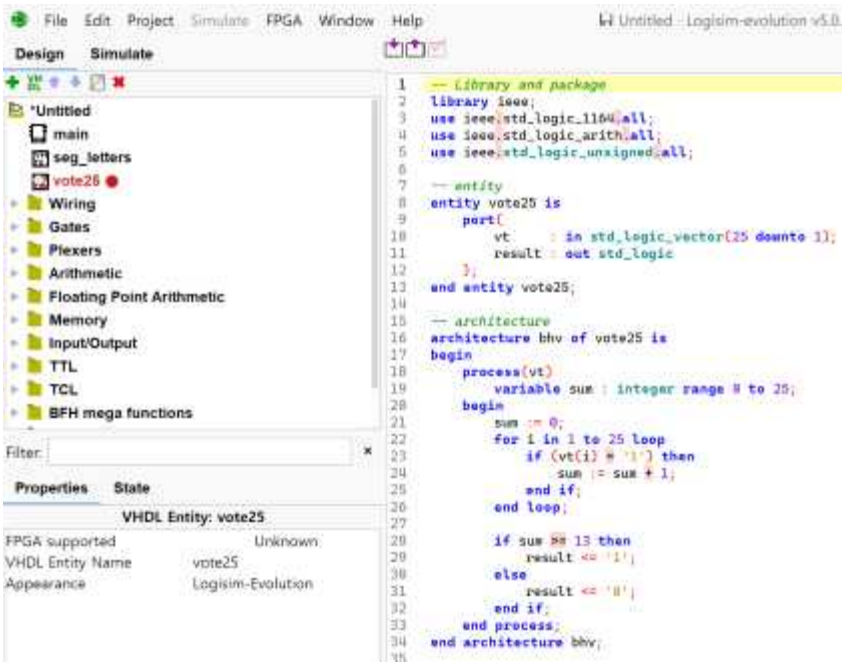
  투표자 입력(찬:1, 반:0) → 25bit vector → vt(25 downto 1)

  → 찬성(1) >= 13 → Output=1

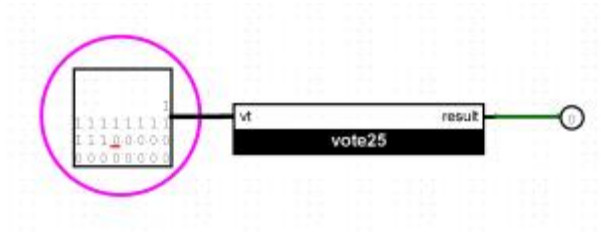
  → 찬성(1) < 13 → Output=0

- 지역 변수와 for-loop 반복문 사용 → 1의 개수 누적 합산 → 과반수 판별

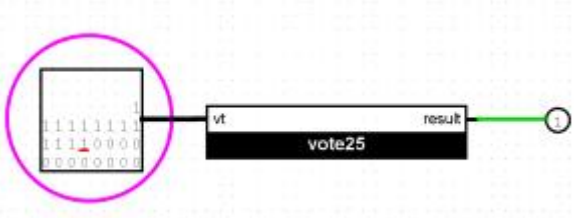
(1) VHDL code (vote25.vhd)



(2) Entity & Simulation (vote25.circ)



(1) < 13, result=0



(1) >= 13, result=1

5. 13bit '1' counter & 7-Segment display (cnt13_seg)

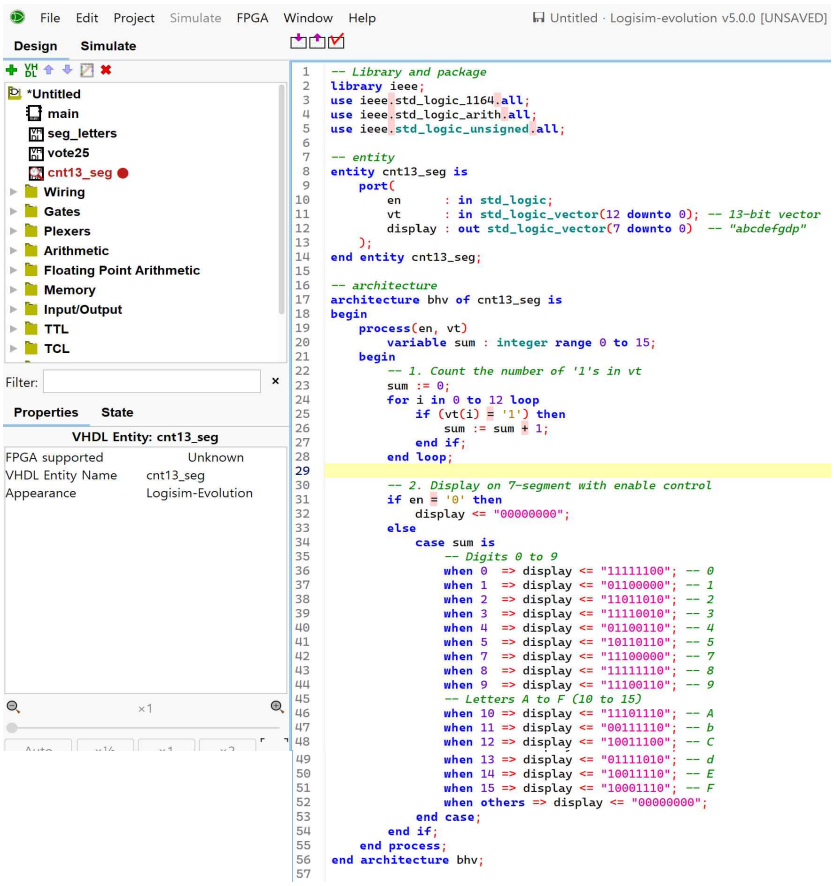
- Input: 13bit 입력 신호 → vt(13 downto 1) → '1'의 개수 count (cnt)
- Output: 7-segment Display

  0 <= cnt <= 9 → 해당 숫자(0~9) 표시

  10 <= cnt <= 15 → 16진수 알파벳 표시 (A~E)

- en=0 → Display off, en=1 → Display output on

(1) VHDL code (cnt13_seg.vhd)




```vhdl
-- Library and package
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

-- entity
entity cnt13_seg is
    port(
        en      : in std_logic;
        vt      : in std_logic_vector(12 downto 0); -- 13-bit vector
        display : out std_logic_vector(7 downto 0)  -- "abcdefgdp"
    );
end entity cnt13_seg;

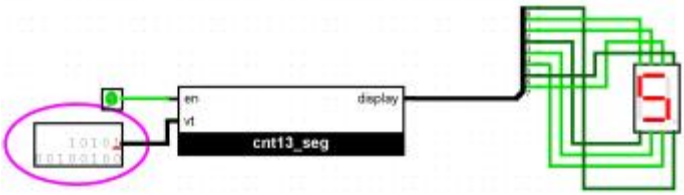
-- architecture
architecture bhv of cnt13_seg is
begin
    process(en, vt)
        variable sum : integer range 0 to 15;
    begin
        -- 1. Count the number of '1's in vt
        sum := 0;
        for i in 0 to 12 loop
            if (vt(i) = '1') then
                sum := sum + 1;
            end if;
        end loop;

        -- 2. Display on 7-segment with enable control
        if en = '0' then
            display <= "00000000";
        else
            case sum is
                -- Digits 0 to 9
                when 0  => display <= "11111100"; -- 0
                when 1  => display <= "01100000"; -- 1
                when 2  => display <= "11011010"; -- 2
                when 3  => display <= "11110010"; -- 3
                when 4  => display <= "01100110"; -- 4
                when 5  => display <= "10110110"; -- 5
                when 7  => display <= "11100000"; -- 7
                when 8  => display <= "11111110"; -- 8
                when 9  => display <= "11100110"; -- 9
                -- Letters A to F (10 to 15)
                when 10 => display <= "11101110"; -- A
                when 11 => display <= "00111110"; -- b
                when 12 => display <= "10011100"; -- C
                when 13 => display <= "01111010"; -- d
                when 14 => display <= "10011110"; -- E
                when 15 => display <= "10001110"; -- F
                when others => display <= "00000000";
            end case;
        end if;
    end process;
end architecture bhv;
```

(2) Entity & Simulation (cnt13_seg.circ)



en=0, display=off



en=1, display=on